CSS MERCANTILE LAW

T8QA

The Question Answers

Topic 8 · Electronic Transactions and Digital Evidence — every past question answered as a model

WHAT THIS DOCUMENT GIVES YOU

- All seven CSS questions on electronic transactions and digital evidence (2022–2026) answered in full
- An answer map for every answer — the verdict on top, each branch a numbered heading carrying complete statements of law
- The command words and limbs of every question decoded
- Model answers of 1,050–1,400 words, in plain English, read against the official texts of the Electronic Transactions Ordinance 2002, the Qanun-e-Shahadat Order 1984 and PECA 2016, with the Supreme Court's digital-evidence cases
- Why each answer scores — what the examiner rewards

Federal Public Service Commission · CSS Competitive Examination · Optional paper, 100 marks

Written for CE-2027 onwards · Record: CE-2016 to CE-2026 · Facts current to October 2026

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HOW TO BUILD THE CONTENTS

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If it does not ask: right-click the grey contents area → Update Field → Update entire table → OK. On a Mac, Control-click the same area. Or press Ctrl + A, then F9 (Windows), to update every field at once. Repeat after any edit, before you export to PDF.

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How to Use These Answers

What is in this document

Table 1 · The questions answered for Topic 8, in the order they were set.

No.	Question	Year · Q	Family
1	What are the functions of the Certification Council under the Electronic Transactions Ordinance, 2002?	CSS 2022 · Q6	The Certification Council and certification service providers
2	What considerations have been provided by the Electronic Transactions Ordinance, 2002 for the purpose of recognition and appraisal of an ele…	CSS 2023 · Q5	Recognition and presumption of electronic documents and signatures
3	What are the powers and functions of the Certification Council under the Electronic Transactions Ordinance, 2002?	CSS 2024 · Q6	The Certification Council and certification service providers
4	Write a detailed note on recognition and presumption of electronic transactions under the Electronic Transactions Ordinance, 2002.	CSS 2024 · Q7	Recognition and presumption of electronic documents and signatures
5	With the rise of digital communication and e-commerce, the Electronic Transactions Ordinance, 2002 has played a crucial role in shaping the …	CSS 2025 · Q7	The Ordinance as a whole — e-contracts, security and enforcement; Recognition and presumption of electronic documents and signatures; The Certification Council and certification service providers
6	Critically examine the presumptions provided under the Electronic Transactions Ordinance, 2002 regarding the authenticity and integrity of e…	CSS 2026 · Q6	Recognition and presumption of electronic documents and signatures
7	Critically analyze the admissibility of electronic and computer-generated evidence under the Qanun-e-Shahadat Order, 1984. How does the judg…	CSS 2026 · Q8	Electronic evidence under the Qanun-e-Shahadat Order

How every answer is built

1. **The question**, exactly as set, with its year, number and marks.
2. **The answer map** — the verdict on top, the question in its own words in the centre (key terms in capitals), and one branch for every numbered heading, each carrying two or three complete statements of law. The **map alone answers the question**.

3. **Decoding the question** — the command words and the limbs, with how to divide the time.
4. **The model answer** — about 1,100–1,300 words; the introduction gives the **statutory definition with its section** and **the answer up front**; every heading carries a section, case or illustration; every answer has a figure besides the map.
5. **Why this answer scores** — what the examiner rewards in it.

EXAMINER'S EYE · HOW THE EXAMINER READS AN ELECTRONIC-TRANSACTIONS ANSWER

This is the **newest** topic — every question dates from 2022 or later — and the examiner expects precise sections. Quote the **definitions** (s. 2(1)(d) advanced signature, (n) electronic signature, (f) authenticity, (q) integrity), the **recognition** sections (ss. 3–7) and the **presumption** (s. 9) in their own words; for the **Council**, give the five members and their qualifications (ss. 18–19) and the nine functions of **s. 21(2)**. Two linked laws earn marks: the **Qanun-e-Shahadat Order** as amended by the Ordinance's **Schedule** (Arts. 30, 46-A, 59, 73, 78-A, 85(6)) and **Article 164** as substituted in **2023**; and **PECA 2016**. For the critical questions, use evidence — low uptake of accredited signatures (NIFT for banks), the PECA overlap, the missing data-protection law, and the Supreme Court's rules on recordings (2016 SCMR 2084; PLD 2019 SC 675).

Answer 1 · The Functions of the Certification Council (CSS 2022)

AT A GLANCE · CSS 2022 · Q6 · 20 MARKS

What are the functions of the Certification Council under the Electronic Transactions Ordinance, 2002?

The answer map

The Certification Council (Electronic Certification Accreditation Council) is a five-member body corporate set up under s. 18 to make digital signatures trustworthy. Its functions under s. 21 are to grant and renew accreditation, monitor accredited providers and suspend or revoke their accreditation, establish and manage the public repository, carry out research, recognise foreign providers, encourage uniform standards, advise any person and make recommendations to authorities. It performs them exclusively (s. 22), through approved practice statements, regulations and speaking orders within ninety days.



Figure 1 · The answer map: every branch is one numbered heading of the model answer below.

Decoding the question

"**What are the functions**" — a **list-and-explain** question. Start with the Council's establishment and composition briefly (ss. 18–19), then explain each function in **s. 21(2)** (a)–(i) with its supporting

provisions (ss. 22–27, 43–44) and an example, and end with how the functions make electronic transactions trustworthy.

The model answer

1. Introduction — the guardian of digital trust

i) Definition

"**Certification Council**" means "the **Electronic Certification Accreditation Council** established under Section 18" (**s. 2(1)(i)**, Electronic Transactions Ordinance 2002). "The Certification Council shall perform such functions as are specified in this Ordinance or may be prescribed" (**s. 21(1)**).

ii) The argument

A digital signature is trustworthy only if the **certification service provider** that vouches for the signer is reliable. The Council is the regulator that **accredits** and **supervises** these providers, keeps a **public repository** of certificates, sets **standards** and **advises**. Its functions make possible the **presumption** of authenticity that the Ordinance gives to accredited advanced signatures (s. 9).

2. Establishment and status

i) Establishment

"Within **sixty days** of the promulgation of this Ordinance, the Federal Government shall, by notification in the official Gazette, constitute a Certification Council to be known as **Electronic Certification Accreditation Council**" (s. 18(1)).

ii) Status

It is "a **body corporate** with **perpetual succession** and a **common seal**", able to **sue and be sued** (s. 18(2)). It operates under the **Ministry of Information Technology and Telecommunication** and is known as **ECAC**.

3. Composition and qualifications

i) Members

Five members, **four** from the **private sector**, one designated chairman (s. 18(3)); appointed by the Federal Government for **three years**, re-appointable **once** (s. 18(4)); removable only for **misconduct** (s. 18(7)); with no **financial interest** in cryptography businesses (s. 18(8)); deciding by **majority**, the chairman having a **casting vote** (s. 18(9)).

ii) Qualifications — s. 19

A **telecommunications engineer** with seven years' experience including one in **cryptography**; **two** IT professionals or academics with seven years; an **administrator** with seven years; and an **advocate** with seven years and knowledge of **IT and telecom law**.

4. Function 1 — granting and renewing accreditation

i) The function

To "**grant and renew accreditation certificates** to certification service providers, their **cryptography services** and **security procedures**" (s. 21(2)(a)).

ii) How it is exercised

The Council accredits a provider, its cryptography services, its electronic or advanced signatures and security procedures that meet the **criteria in its regulations** (s. 24(1)); terms, duration and renewal are set by regulations (s. 24(2)); fees are prescribed (s. 24(3)); and the procedure must be **transparent** "with due regard to the **right of hearing**" (s. 24(4)). A provider must also submit its **certification practice statement** for approval (s. 25(3)).

iii) Illustration

SecureSign applies for accreditation. The Council checks its identity-verification process, key security and staff qualifications, approves its practice statement, and grants accreditation. Ms Sara's certificate from SecureSign then supports an **advanced** signature presumed genuine (s. 9).

5. Function 2 — monitoring, suspension and revocation

i) The function

To "**monitor and ensure compliance** by accredited certification service providers with the terms of their accreditation and **revoke or suspend** accreditation" (s. 21(2)(b)), and to "**monitor compliance** … with the provisions of this Ordinance" (s. 21(2)(c)).

ii) Due process

Suspension or revocation for non-compliance with the practice-statement rules may be ordered only "after providing **reasonable right of hearing**" (s. 25(7)), and notice must be posted in the **repository** (s. 23(4)).

6. Function 3 — the repository

i) The function

To "**establish and manage the repository**" (s. 21(2)(d)) — a database of all **accreditation certificates**, certificates issued by **accredited providers**, and other specified information (s. 23(1)).

ii) Its features

The Council must keep it **secure** (s. 23(2)); "all information contained in the repository shall be **open to public inspection**" (s. 23(3)); notices of **suspension or revocation** must be posted within the prescribed time (s. 23(4)). Its certificates are **public documents** (QSO Art. 85(6)), and a fee of up to **ten rupees** may be charged for each certificate deposited (s. 20(c)).

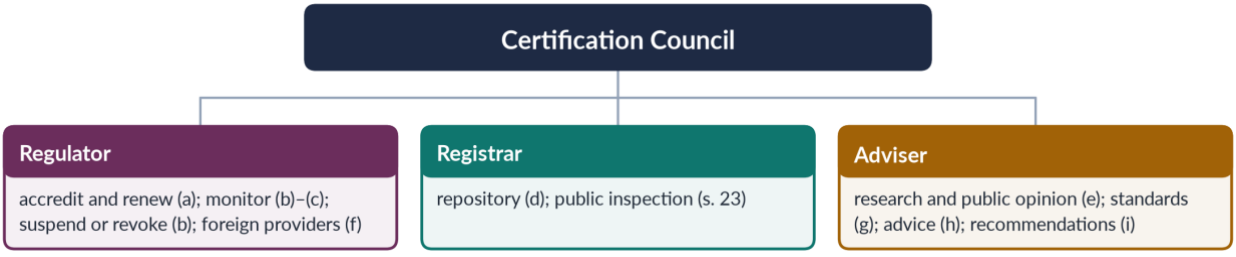


Figure 2 · The functions of the Certification Council (s. 21(2)).

7. Function 4 — research and public opinion

i) The function

To "carry out **research and studies** in relation to cryptography services and to obtain **public opinion** in connection therewith" (s. 21(2)(e)).

ii) Public consultation

All draft **rules and regulations** must be published in the Gazette and in an **English and an Urdu daily** at least **thirty days** before coming into force; the Council must record every comment and publish a **report** answering them (s. 44).

8. Function 5 — foreign certification service providers

i) The function

To "**recognize or accredit foreign** certification service providers" (s. 21(2)(f)).

ii) Why it matters

Cross-border trade needs foreign signatures to be trusted in Pakistan; regulations may provide for **cross-certification**, recognition and **bridge** arrangements with providers abroad (s. 43(2)(g)). If Gulf Garments in Dubai signs with a certificate from a recognised foreign provider, Pakistani courts can rely on it.

9. Function 6 — standards, advice and recommendations

i) Standards

To "**encourage uniformity of standards and practices**" (s. 21(2)(g)), supported by regulations on protocols, algorithms, certificate profiles and audit trails (s. 43(2)(k), (o)–(s)).

ii) Advice and recommendations

To "give **advice** to any person in relation to any matter covered under this Ordinance" (s. 21(2)(h)) and to make **recommendations** to an **appropriate authority** (s. 21(2)(i)) — for instance, on using accredited signatures in government e-services under s. 16.

10. Supporting powers — exclusivity, practice statements, regulations

i) Exclusive power

"Notwithstanding anything contained in the Pakistan Telecommunication (Re-organisation) Act, 1996", the Council is "**exclusively responsible**" to grant, renew, suspend or revoke accreditation (s. 22).

ii) Practice statements and regulations

It approves each provider's **certification practice statement** and its changes (s. 25(3)–(4)), and makes **regulations**, with the Federal Government's approval, on keys, time-stamping, staff qualifications, data privacy, inspections and reparation for negligence (s. 43).

11. How the functions are performed — transparency

i) Speaking orders

"All applications and matters coming before the Certification Council shall be decided through a **speaking order**, as expeditiously as possible but **not later than ninety days**" except in extraordinary circumstances with recorded reasons (s. 26).

ii) Resources

Funds come from **Government grants**, accreditation **fees**, the repository fee and **fines** (s. 20); it may appoint officers and advisers and open regional offices (s. 27). Accreditation decisions **cannot be delegated** to individual members (s. 18(6)).

12. Conclusion — a regulator for digital trust

i) Summary

The Council's functions are to **accredit** and **renew**; **monitor**, **suspend** and **revoke**; keep the **repository**; **research** and consult; recognise **foreign** providers; encourage **standards**; **advise**; and **recommend** (s. 21(2)) — exercised **exclusively** (s. 22), through practice statements, regulations and speaking orders (ss. 25–26, 43–44).

ii) The verdict

These functions make accredited signatures reliable enough for the law to **presume** them genuine (s. 9). In practice the Council has accredited institutions such as **NIFT**, which serves banks; its wider impact depends on making accredited certificates common in business and government.

Why this answer scores

- It sets the functions in context with the Council's establishment and expert composition.
- Each function of s. 21(2) is explained with its supporting sections.
- The repository, exclusivity and practice-statement powers show depth.
- Transparency rules (speaking orders, public consultation) are highlighted.
- The tree figure groups the nine functions into three roles.

Answer 2 · Considerations for the Recognition and Appraisal of an Electronic Document (CSS 2023)

AT A GLANCE · CSS 2023 · Q5 · 20 MARKS

What considerations have been provided by the Electronic Transactions Ordinance, 2002 for the purpose of recognition and appraisal of an electronic document?

The answer map

The Ordinance recognises an electronic document unless something other than its electronic form defeats it (s. 3), and appraises it by functional tests — accessibility for writing, reliable integrity and legibility for originals, accuracy and metadata for retention (ss. 4–6). Signatures may be proved in any manner, and advanced ones are presumed genuine (ss. 7–9); attribution and timing rules fix whose document it is (ss. 13–15); and the Qanun-e-Shahadat amendments make printouts primary evidence and require proof of a security procedure when authorship is denied.



Figure 3 · The answer map: every branch is one numbered heading of the model answer below.

Decoding the question

"**What considerations**" — the **tests** the Ordinance uses. "**Recognition**" — whether the document counts in law (ss. 3–7, 10–12, 30); "**appraisal**" — how reliable it is and how it is proved (ss. 5, 8–9, 13–15; QSO

Arts. 46-A, 73, 78-A). Organise the answer as a list of considerations, each with its section, and end with a short table.

The model answer

1. Introduction — when the law trusts an electronic document

i) Definition

"**Electronic document**" includes "documents, records, information, communications or transactions in electronic form" (**s. 2(1)(m)**, Electronic Transactions Ordinance 2002). "No document … shall be **denied legal recognition**, admissibility, effect, validity, proof or enforceability on the ground that it is in **electronic form**" (**s. 3**).

ii) The argument

The Ordinance applies **functional equivalence**: an electronic document is recognised wherever it performs the **function** of the paper rule — later reference, integrity, identification of the signer. **Appraisal** then turns on **integrity**, **security procedures** and **certification**, and on the evidence rules the Ordinance added to the Qanun-e-Shahadat Order.

2. Non-discrimination — s. 3

i) The rule

Electronic form, or the absence of an **attesting witness**, is **no ground** to deny a document recognition, admissibility, effect, validity, proof or enforceability.

ii) Its limit

Section 3 removes only **that** objection. A document can still fail for other reasons — lack of agreement, fraud, or a defect in authenticity.

3. Writing — accessibility (s. 4)

i) The test

A legal requirement of **writing** is satisfied by an electronic document "if the same is **accessible so as to be usable for subsequent reference**".

ii) Illustration

Gulf Garments' acceptance email, saved in Sara Fabrics' mailbox and printable at any time, is "writing". A voice call that was never recorded, or a disappearing message, is not.

4. Original form — integrity and legibility (s. 5)

i) Integrity

There must be "a **reliable assurance as to the integrity**" of the document "from the time when it was first generated in its **final form**" (s. 5(1)(a)) — that it has "remained **complete and unaltered**", apart from normal endorsements or changes in communication, storage or display (s. 5(2)(a)).

ii) Reliability and legibility

The standard of reliability depends on the **purpose** and **all relevant circumstances** (s. 5(2)(b)); and the document must be capable of being **displayed in a legible form** (s. 5(1)(b)).

5. Retention — accuracy and metadata (s. 6)

Where a law requires retention, an electronic record suffices if (a) it remains **accessible**; (b) its contents and form are **as originally generated, sent or received**, or accurately represent them; and (c) information identifying its **origin and destination** and the **date and time** of generation, sending or receipt is retained. These **metadata** are often decisive in appraising authenticity.

If Imran forwards Gulf's email to himself and edits the price before printing it, the edited copy fails s. 6(b); the original in the mail server, with its header showing sender, time and route, is the reliable record.

6. Signature — electronic and advanced (ss. 7–8)

i) Recognition

A requirement of **signature** is satisfied by an **electronic signature** or **advanced electronic signature** (s. 7). An electronic signature is any letters, numbers, symbols or images applied "**with the intention of authenticating or approving**" the document (s. 2(1)(n)).

ii) Proof

"An electronic signature **may be proved in any manner**" to verify that the document is of the person who executed it (s. 8) — witnesses, account records, admissions, forensic evidence.

7. The presumption for advanced signatures (s. 9)

i) What is presumed

In proceedings involving an **advanced** electronic signature, it is presumed — "unless evidence to contrary is adduced" — that the document identified in a **valid accreditation certificate** is **authentic** and has **integrity**, that the signature is that of the person to whom it **correlates**, that he affixed it **intending to sign**, and that the document **has not been altered** (s. 9).

ii) Its significance for appraisal

A document with an accredited **advanced** signature starts with a **statutory presumption**; one with an ordinary signature must be proved.

Table 2 · The considerations at a glance.

Consideration	Section	Test
Non-discrimination	3	Not rejected for being electronic or unwitnessed
Accessibility	4	Usable for later reference
Integrity and legibility	5	Complete, unaltered, readable
Accuracy and metadata	6	As generated; origin, destination, time kept
Signature	7–8	Intent to authenticate; proof in any manner
Presumption	9	Advanced signature presumed genuine
Attribution and timing	13–15	Whose document; when and where effective
Evidence	QSO 73, 78-A	Primary evidence; security procedure if denied

8. Attribution and communication (ss. 13–15)

i) Whose document

A communication is the **originator's** if sent by **him**, by a person **authorised** to act for him, or by an **automated system programmed** by or for him (s. 13(1)); the addressee may rely on it unless he **suspects**, or **ought to know** by reasonable care, that it is not authentic (s. 13(2)).

ii) When and where

Dispatch occurs when it enters a system **outside the originator's control**; **receipt** on entry into the addressee's designated (or any) system, or on retrieval from a non-designated one; the **place** is the parties' place of business (s. 15).

9. Formalities — stamp, attestation, certified copies (ss. 10–12)

i) Stamp duty and attestation

No stamp duty (s. 10) and **no attestation or notarisation** (s. 11) for two years or until the Provinces or authorities create electronic systems, **whichever is later**.

ii) Certified copies

Requirements for **certified copies** extend to **printouts** verified as the **appropriate authority** lays down (s. 12).

10. Appraisal in evidence — the Qanun-e-Shahadat amendments

i) Relevance and primary evidence

Records of **automated systems** working properly are **relevant facts** (Art. 46-A); a **printout** of an automated system is **primary evidence**, its working order **presumed** (Art. 73, Expl. 3); a printout of another electronic document is primary evidence where a **security procedure** was applied (Art. 73, Expl. 4).

ii) Denied authorship

"Where such allegation is **denied**, the **application of a security procedure** to the signature or the electronic document **must be proved**" (Art. 78-A). **Experts** may testify on authenticity and integrity (Art. 59).

11. Limits — excluded instruments and judicial appraisal

i) Exclusions

The Ordinance does not apply to **negotiable instruments**, **powers of attorney**, express **trusts**, **wills** and contracts for the **sale of immovable property**, unless extended by the Government (s. 31).

ii) Judicial weight

Recognition does not guarantee **weight**: courts insist that recordings and digital records be proved **genuine and untampered** (*Ishtiaq Ahmed Mirza v Federation of Pakistan*, **PLD 2019 SC 675**).

12. Conclusion — functional equivalence with safeguards

i) Summary

The Ordinance recognises electronic documents by **non-discrimination** (s. 3) and **functional tests** — accessibility (s. 4), integrity and legibility (s. 5), accuracy and metadata (s. 6), electronic signature (s. 7) — and appraises them through **proof in any manner** (s. 8), the **presumption** for advanced signatures (s. 9), **attribution and timing** (ss. 13–15) and the **QSO** rules (Arts. 46-A, 59, 73, 78-A).

ii) The verdict

The considerations put electronic documents on an **equal footing** with paper where they serve the same purposes, while **integrity, security procedures and certification** decide how much they prove.

Why this answer scores

- It distinguishes recognition from appraisal, as the question invites.
- Each consideration is tied to its section, with the statute's words.
- The QSO amendments complete the appraisal limb.
- The table summarises all the considerations at a glance.
- Exclusions and judicial weight show the limits.

Answer 3 · The Powers and Functions of the Certification Council (CSS 2024)

AT A GLANCE · CSS 2024 · Q6 · 20 MARKS

What are the powers and functions of the Certification Council under the Electronic Transactions Ordinance, 2002?

The answer map

The Certification Council is an independent, expert body corporate whose powers give teeth to its functions. It has the exclusive power to grant, renew, suspend and revoke accreditation, to approve providers' practice statements, to make regulations with Government approval and to run the public repository; its functions also include research, recognition of foreign providers, standards, advice and recommendations. It must act transparently — fair hearings, public consultation and speaking orders within ninety days — and is funded by grants, fees and fines.



Figure 4 · The answer map: every branch is one numbered heading of the model answer below.

Decoding the question

Two words: "**powers**" (what the Council **can do** — ss. 18(6), 22–27, 43–44) and "**functions**" (what it **must do** — s. 21). Give the powers the larger share, since the 2022 question already asked for functions, and show how each power serves a function. Begin with status and composition, and end with a brief assessment.

The model answer

1. Introduction — powers that make functions work

i) Definition

The Federal Government shall "constitute a Certification Council to be known as **Electronic Certification Accreditation Council**" (**s. 18(1)**, Electronic Transactions Ordinance 2002), which "shall perform such **functions** as are specified in this Ordinance or may be prescribed" (**s. 21(1)**).

ii) The argument

The Council's **functions** — accreditation, supervision, repository, research, standards, advice — are listed in s. 21. Its **powers** — exclusive accreditation, suspension and revocation, approval of practice statements, regulations, control of the repository, speaking orders — give those functions **teeth**. Together they create the **trust** on which the presumption for advanced signatures (s. 9) rests.

2. Legal status and composition

i) Status

A **body corporate** with perpetual succession and a common seal, which may **sue and be sued** (s. 18(2)).

ii) Composition

Five members, **four** from the **private sector**, one designated chairman (s. 18(3)); appointed by the Federal Government for **three years**, re-appointable **once** (s. 18(4)); qualified as a **telecom engineer** with cryptography experience, **two IT** professionals, an **administrator** and an **advocate**, each with **seven years'** experience (s. 19).

3. Independence and decision-making

i) Tenure and integrity

A member "shall not be removed **except on the grounds of misconduct**" (s. 18(7)) and may have no **direct financial interest** in cryptography businesses (s. 18(8)).

ii) Decisions

Decisions are by **majority**, with the chairman's **casting vote** (s. 18(9)); the Council may delegate functions to members, **except** the **grant, renewal, revocation or suspension of accreditation** (s. 18(6)); no act is invalid merely because of a vacancy or defect in constitution (s. 18(5)).

4. Power to accredit — exclusive

i) Exclusivity

"Notwithstanding anything contained in the Pakistan Telecommunication (Re-organisation) Act, 1996", the Council is "**exclusively responsible** to grant, renew, suspend or revoke the accreditation to certification service providers, their cryptography services and security procedures" (s. 22).

ii) Grant

It "may grant accreditation to certification service provider, its cryptography services, electronic signature or advanced electronic signature and security procedures who complies with the **criteria for accreditation specified in the regulations**" (s. 24(1)), on terms set by regulations (s. 24(2)) and for prescribed fees (s. 24(3)).

iii) Effect

Only an accredited provider may **hold itself out** as accredited (s. 17(2)); its signatures may qualify as **advanced** (s. 2(1)(d)(ii)) and attract the **presumption** of s. 9.

5. Power to suspend and revoke

i) The power

The Council must "**monitor and ensure compliance** … and **revoke or suspend** accreditation in the manner and on the grounds … specified in the regulations" (s. 21(2)(b)), and may suspend or revoke for failure to comply with the practice-statement rules (s. 25(7)).

ii) The safeguard

The procedure must be "**transparent** … with due regard to the **right of hearing**" (s. 24(4)), and suspension or revocation follows "**reasonable right of hearing**" (s. 25(7)).

iii) Illustration

SecureSign issues certificates without proper identity checks. The Council hears it, **suspends** its accreditation, and posts the suspension in the **repository** (s. 23(4)); certificates issued while suspended expose its officers to prosecution (s. 35(1)(d)).

6. Power over practice statements

Every provider seeking accreditation must prepare a **certification practice statement** in the form specified by regulations (s. 25(1)), covering subscriber identification, revocation, accuracy and reliance (s. 25(2)), and **submit it for approval** (s. 25(3)); changes require the Council's **approval** (s. 25(4)); a copy is kept at the Council, **open to public inspection** (s. 25(5)). During accreditation, the provider is **deemed to warrant** compliance and accuracy (s. 25(6)).

Table 3 · Powers and functions of the Certification Council.

Power or function	Section	Content
Exclusive accreditation	22, 24	Grant and renew; criteria in regulations; hearing
Suspension and revocation	21(2)(b), 25(7)	After monitoring and a reasonable hearing
Practice statements	25	Approve, control changes, public inspection
Regulations	43–44	With Government approval; 30-day consultation
Repository	21(2)(d), 23	Establish, secure, public, post revocations
Research, foreign, standards	21(2)(e)–(g)	Studies; recognise foreign providers; uniformity
Advice and recommendations	21(2)(h)–(i)	Any person; appropriate authorities
Decisions and resources	20, 26–27	Speaking orders in 90 days; grants, fees, fines; staff

7. Power to make regulations

i) Scope

With the Federal Government's **prior approval**, the Council may make regulations on the **safety of keys and passwords**, **time-stamping**, **staff qualifications**, **facilities**, **data privacy** of subscribers, **inspection**, **cross-certification** with foreign providers, **reparation** for negligence, certificate and revocation-list **profiles**, **record retention** and **audit trails** (s. 43).

ii) Consultation

Drafts must be published in the Gazette and in an **English and an Urdu daily** at least **thirty days** in advance, and the final regulations must be accompanied by a **report** answering every comment (s. 44).

8. Power over the repository

The Council establishes and manages a **repository** of accreditation certificates and certificates issued by accredited providers (s. 23(1)), keeps it **secure** (s. 23(2)), keeps it **open to public inspection** (s. 23(3)), and posts notices of **suspension or revocation** (s. 23(4)). Repository certificates are **public documents** (QSO Art. 85(6)), and a fee of up to **ten rupees** per certificate may be charged (s. 20(c)).

9. Functions — research, foreign providers and standards

i) Research

To carry out **research and studies** on cryptography services and obtain **public opinion** (s. 21(2)(e)).

ii) Foreign providers and standards

To **recognise or accredit foreign** providers (s. 21(2)(f)) — enabling cross-border trust — and to **encourage uniformity** of standards and practices (s. 21(2)(g)).

10. Functions — advice and recommendations

To give **advice to any person** on matters under the Ordinance (s. 21(2)(h)) and make **recommendations** to an **appropriate authority** (s. 21(2)(i)) — for example, on the signatures and security procedures a government department should require for e-filing under s. 16.

11. Procedural and financial powers

i) Speaking orders

All matters are decided by a **speaking order**, "not later than **ninety days**" except in extraordinary circumstances with recorded reasons (s. 26).

ii) Staff and funds

It may appoint **officers, employees and advisers** and open **regional offices** (s. 27). Its funds are **Government grants**, accreditation **fees**, the repository fee and **fines** (s. 20).

12. Conclusion — a well-armed but underused regulator

i) Summary

The Council's **functions** (s. 21) are performed through its **powers**: exclusive accreditation (ss. 22, 24), suspension and revocation after hearing (ss. 21(2)(b), 25(7)), control of practice statements (s. 25), regulations after consultation (ss. 43–44), the public repository (s. 23), and speaking orders within ninety days (s. 26), backed by an expert, protected membership (ss. 18–19).

ii) The verdict

The design is **sound and transparent**. In practice the Council has accredited institutions such as **NIFT** for the banking sector, but accredited signatures remain **rare** among citizens and small businesses; its powers will matter more as digital identity and e-government expand.

Why this answer scores

- It separates powers from functions and gives the powers the larger share.
- Each power is cited with its section and its link to a function.
- Due-process safeguards — hearing, consultation, speaking orders — are emphasised.
- The table lists every power and function at a glance.
- A brief assessment completes the answer.

Answer 4 · Recognition and Presumption of Electronic Transactions (CSS 2024)

AT A GLANCE · CSS 2024 · Q7 · 20 MARKS

Write a detailed note on recognition and presumption of electronic transactions under the Electronic Transactions Ordinance, 2002.

The answer map

Chapter 2 of the Ordinance (ss. 3–12) recognises electronic documents, writing, originals, retention and signatures on functional tests, suspends stamp duty and attestation until electronic systems exist, and allows certified printouts. Its presumption (s. 9) treats an advanced electronic signature as authentic, intended and unaltered unless contrary evidence is adduced, and the Qanun-e-Shahadat amendments add presumptions that automated systems were working and printouts are primary evidence.



Figure 5 · The answer map: every branch is one numbered heading of the model answer below.

Decoding the question

"**Write a detailed note**" on "**recognition and presumption**" — essentially **Chapter 2** of the Ordinance (ss. 3–12), with s. 30 and the related **QSO** presumptions. Explain each section with its words and an example; give the **presumption** (s. 9) its own full heading, and add the evidential presumptions of Arts. 73 and 78-A.

The model answer

1. Introduction — recognition and presumption as twin pillars

i) Definition

"No document, record, information, communication or transaction shall be **denied legal recognition, admissibility, effect, validity, proof or enforceability** on the ground that it is in **electronic form** and has not been attested by any witness" (**s. 3**, Electronic Transactions Ordinance 2002). In proceedings involving an advanced electronic signature, its authenticity and integrity "**shall be presumed unless evidence to contrary is adduced**" (**s. 9**).

ii) The argument

Recognition answers whether an electronic transaction **counts** in law; **presumption** answers how easily it can be **proved**. Chapter 2 of the Ordinance (ss. 3–12) provides both, using **functional equivalence** and a statutory presumption for **advanced** signatures.

2. Recognition of electronic forms — s. 3

i) Seven protections

Electronic form cannot be the reason to deny a document **recognition, admissibility, effect, validity, proof or enforceability**; nor can the absence of an **attesting witness**.

ii) Illustration

Gulf Garments cannot avoid its email acceptance merely by arguing that it was "only an email" or that no one witnessed it.

iii) Its limit

Section 3 removes only the objection of **electronic form**. The document may still fail for other reasons — no real agreement, fraud, or inability to prove who sent it.

3. Writing — s. 4

A legal requirement for writing is "deemed satisfied where the document … is in electronic form, if the same is **accessible so as to be usable for subsequent reference**". The test is **accessibility**, not paper. An arbitration clause contained in an exchange of emails satisfies the written-agreement requirement of the Arbitration Act if the emails are saved.

4. Original form — s. 5

i) Conditions

A requirement to present or retain an **original** is met if there is "**a reliable assurance as to the integrity**" of the document since it was first generated in **final form**, and it can be **displayed legibly** (s. 5(1)).

ii) Tests

Integrity means remaining "**complete and unaltered**", apart from normal endorsements and changes in communication, storage or display (s. 5(2)(a)); reliability depends on the **purpose** and **circumstances** (s. 5(2)(b)).

5. Retention — s. 6

Retention in electronic form suffices if the record is **accessible**, its contents and form are **as originally generated, sent or received** (or accurately represent them), and information on its **origin, destination, date and time** is retained (s. 6(a)–(c)).

6. Recognition of electronic signatures — s. 7

i) The rule

"The requirement under any law for **affixation of signatures** shall be deemed satisfied where **electronic signatures or advanced electronic signature** are applied" (s. 7).

ii) What counts

An **electronic signature** is any letters, numbers, symbols, images or characters in electronic form applied "**with the intention of authenticating or approving**" the document (s. 2(1)(n)) — a typed name, a scanned signature, a click on "I agree". An **advanced** signature is unique, identifying, under the signer's **sole control** and makes later changes **detectable**, or is accredited by the Council (s. 2(1)(d)).

7. Proof of electronic signatures — s. 8

"An electronic signature **may be proved in any manner**, in order to verify that the electronic document is of the person that has executed it with the intention and for the purpose of verifying its authenticity or integrity or both" (s. 8). Proof may come from email account records, IP logs, witnesses, admissions or forensic reports. If authorship is **denied**, the party relying on the document must prove that a **security procedure** was applied (QSO Art. 78-A).

8. The presumption — s. 9

i) What is presumed

In any proceedings involving an **advanced electronic signature**, it is presumed, **unless evidence to contrary is adduced**, that (a) the document identified in a **valid accreditation certificate** is **authentic and has integrity**; or (b) the signature is that of the person to whom it **correlates**, was affixed by him **with the intention of signing or approving**, and the document **has not been altered** since then.

ii) Its nature

The presumption is **rebuttable** and available **only** for advanced signatures; it shifts to the denying party the burden of producing evidence — of a **stolen key**, a **revoked** certificate or **tampering**.

iii) Illustration

Ms Sara's export contract signed with Gulf Garments' **SecureSign** certificate is presumed signed by Gulf and unaltered; a contract bearing only Gulf's typed name must be proved under s. 8.

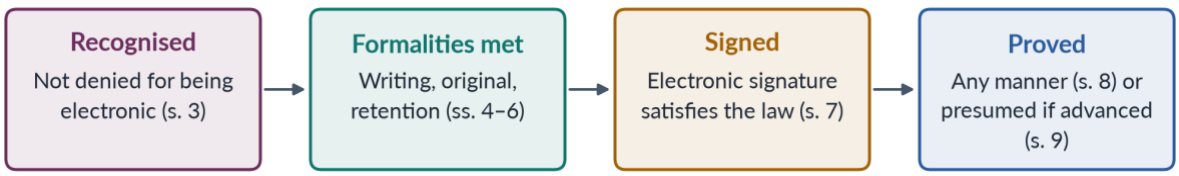


Figure 6 · Recognition and presumption, step by step.

9. Stamp duty and attestation — ss. 10–11

i) Stamp duty

"Notwithstanding anything contained in the **Stamp Act, 1899**", for **two years** or until the **Provinces** devise and implement measures for electronic payment of stamp duty, **whichever is later**, stamp duty is **not payable** on electronic instruments (s. 10).

ii) Attestation and notarisation

For **two years** or until the **appropriate authority** devises measures for electronic attestation and notarisation, **whichever is later**, no electronic document requires **attestation or notarisation** (s. 11).

10. Certified copies and paper words — ss. 12, 30

i) Certified copies

Requirements for **certified copies** extend to **printouts** or other displays of electronic documents verified in the manner laid down by the **appropriate authority** (s. 12).

ii) Paper words

"Attestation", "document", "original", "record", "seal", "signature", "writing" and other paper words "shall, **mutatis mutandis**, extend to electronic forms" (s. 30), so recognition reaches every statute.

iii) Exclusions

Negotiable instruments, powers of attorney, express trusts, wills and contracts for sale of immovable property are **outside** these rules unless the Government extends them (s. 31).

11. Presumptions in the evidence law

i) Automated systems

A **printout** of an automated information system is **primary evidence** if the system was working, and "in the absence of evidence to the contrary, it shall be **presumed** that the automated information system was in working order" (QSO Art. 73, Expl. 3); its records are **relevant facts** (Art. 46-A).

ii) Other electronic documents

A printout of another electronic document is primary evidence where a **security procedure** was applied (Art. 73, Expl. 4); and where authorship is **denied**, the security procedure "**must be proved**" (Art. 78-A).

12. Conclusion — validity with proof

i) Summary

The Ordinance **recognises** electronic transactions through non-discrimination (s. 3), writing (s. 4), originals (s. 5), retention (s. 6), signatures (s. 7), suspended formalities (ss. 10–11), certified printouts (s. 12) and the extension of paper words (s. 30). It **presumes** advanced signatures genuine (s. 9), and the QSO presumes automated systems were working (Art. 73).

ii) The verdict

Recognition puts electronic transactions **on a par with paper**; the presumption **rewards** secure, accredited signatures. Their full benefit depends on wider adoption of accredited certificates.

Why this answer scores

- It covers the whole of Chapter 2 (ss. 3–12), with s. 30 and s. 31.
- The presumption of s. 9 is quoted and explained with its limits.
- The QSO presumptions add the evidential dimension of "presumption".
- The flow figure and running example make the scheme concrete.
- The conclusion links recognition and presumption to practice.

Answer 5 · The Electronic Transactions Ordinance — Key Provisions, Effectiveness and Challenges (CSS 2025)

AT A GLANCE · CSS 2025 · Q7 · 20 MARKS

With the rise of digital communication and e-commerce, the Electronic Transactions Ordinance, 2002 has played a crucial role in shaping the legal landscape of electronic contracts and digital signatures in Pakistan. Discuss the key provisions of the Ordinance, focusing on the legal recognition of electronic records and signatures, as well as the role of certification authorities. Critically evaluate the effectiveness of this Ordinance in addressing issues of security, authenticity and reliability in digital transactions. What challenges remain in its enforcement, particularly in the context of Pakistan's evolving digital economy?

The answer map

The Ordinance recognises electronic records and signatures, governs electronic contracts and communications, and creates a trust system of accredited certification service providers supervised by the Certification Council, backed by offences and evidence rules. It addresses authenticity and reliability well in law — especially through the presumption for advanced signatures and the Qanun-e-Shahadat amendments — and security partly. Its effectiveness is limited by low use of accredited signatures, open-ended formalities, overlap with PECA, the absence of a data-protection law and weak forensic and institutional capacity.



Figure 7 · The answer map: every branch is one numbered heading of the model answer below.

Decoding the question

A four-part critical question: (1) **key provisions** — recognition of electronic records and signatures; (2) the **role of certification authorities**; (3) **critically evaluate** effectiveness on **security, authenticity and**

reliability; (4) **challenges** in enforcement in the **digital economy**. Give about 40% to the law, 40% to evaluation and 20% to challenges and reforms, with evidence.

The model answer

1. Introduction — the legal foundation of the digital economy

i) Definition

The Electronic Transactions Ordinance 2002 (LI of 2002) was made "to **recognize and facilitate** documents, records, information, communications and transactions in **electronic form**, and to provide for the **accreditation of certification service providers**" (long title). "No document … shall be denied legal recognition … on the ground that it is in electronic form" (**s. 3**).

ii) The argument

The Ordinance gave Pakistan, from 2002, a **sound legal foundation** for e-commerce: recognition of electronic records and signatures, contract rules, a certification system and evidence rules. It addresses **authenticity** and **reliability** well **in law** and **security** partly. Its **effectiveness** has been limited by **low adoption**, **unfinished systems**, **overlapping laws** and **weak capacity**.

2. Legal recognition of electronic records

i) Non-discrimination

Electronic form or lack of an attesting witness cannot deny a document **recognition, admissibility, effect, validity, proof or enforceability** (s. 3).

ii) Functional tests

Writing — accessible for later reference (s. 4); **original** — reliable assurance of **integrity** and legible display (s. 5); **retention** — accessible, accurate, with **origin, destination, date and time** (s. 6). Paper words such as "writing", "signature" and "original" **extend** to electronic forms (s. 30). Stamp duty and attestation are **suspended** until electronic systems exist (ss. 10–11).

3. Legal recognition of electronic signatures

i) Two tiers

Any **electronic signature** — a mark applied **with intent to authenticate** (s. 2(1)(n)) — satisfies a signature requirement (s. 7) and may be **proved in any manner** (s. 8). An **advanced** signature — unique, identifying, under **sole control**, change-detectable, or Council-accredited (s. 2(1)(d)) — is **presumed** authentic, intended and unaltered unless contrary evidence is adduced (s. 9).

ii) In evidence

Where authorship of an electronic document is **denied**, the **security procedure** must be proved (QSO Art. 78-A).

4. Electronic contracts and communications

i) Attribution — s. 13

A message is the **originator's** if sent by him, his **authorised agent** or his **programmed automated system**; the addressee may rely on it unless he suspects or **ought to know** it is not authentic.

ii) Acknowledgment, dispatch and receipt — ss. 14–15

A message conditional on acknowledgment is **unsent** until acknowledged; **dispatch** occurs when it leaves the sender's **control**; **receipt** on entry into the addressee's designated (or any) system; the **place** is the parties' place of business. Together with the **Contract Act**, these rules make **email and online contracts** enforceable.

5. The role of certification authorities

i) Certification service providers

A provider issues **certificates** confirming the **authenticity or integrity** of documents and signatures (s. 2(1)(h)); it may operate unaccredited, but may not claim accreditation without a **valid accreditation certificate** (s. 17). An accredited provider keeps an approved, public **practice statement** and is **deemed to warrant** compliance and accuracy (s. 25).

ii) The Certification Council

The **Electronic Certification Accreditation Council** — five members, four private, with IT, cryptography, administrative and legal expertise (ss. 18–19) — **exclusively** accredits providers (s. 22), **monitors**, **suspends** and **revokes** them after hearing (ss. 21, 25(7)), keeps a public **repository** of certificates (s. 23), recognises **foreign** providers and sets **standards** (s. 21). It has accredited **NIFT**, which issues certificates to banks.

Table 4 · Effectiveness of the Ordinance — security, authenticity, reliability.

Test	Strengths	Weaknesses
Security	Offences (ss. 34–37), up to 7 years; secret keys (s. 41); reach abroad (s. 32)	Overlap with PECA; no data-protection law; enforcement capacity
Authenticity	Attribution (s. 13); s. 9 presumption; accredited certification (ss. 17–25)	Few accredited signatures; hacked and shared accounts
Reliability	Integrity tests (ss. 5–6); printouts primary evidence (QSO Art. 73)	Tampering and deepfakes; forensic gaps; working-order presumption favours institutions

6. Security — offences and safeguards

i) Offences

False information by **subscribers** and false **certificates** by providers — up to **seven years** or **Rs 10 million**, with compensation (ss. 34–35); **unauthorised access** and **damage** to information systems — up to seven years or Rs 1 million (ss. 36–37); all **non-bailable, compoundable and cognizable** (s. 38), tried by the **Court of Sessions** (s. 39).

ii) Safeguards

No one may be compelled to disclose a **secret key** except for offences (s. 41); network providers are not liable without **intent** (s. 40); the Ordinance applies to acts **abroad** affecting Pakistan (s. 32).

7. Evaluating security

i) Strengths

The **certification offences** protect the trust system directly, and the maxima are **heavy**.

ii) Weaknesses

The access and damage offences **overlap** with **PECA 2016**, which punishes unauthorised access with only **three months** (PECA s. 3) and provides the **investigation** powers the Ordinance lacks; and Pakistan had **no comprehensive data-protection law** by the end of 2025, leaving personal data in e-commerce weakly protected.

8. Evaluating authenticity

i) Strengths

Attribution rules (s. 13) and the **presumption** for advanced signatures (s. 9) give clear answers on **who** made a document.

ii) Weaknesses

Accredited signatures are used mainly by **banks** and institutions; citizens and small businesses rely on **emails, scans and WhatsApp**, which get **no presumption** and are vulnerable to **hacked or shared** accounts.

9. Evaluating reliability

i) Strengths

Integrity is the test for originals and retention (ss. 5–6); printouts of automated systems are **primary evidence** with a presumption of **working order**, and expert opinion on integrity is admissible (QSO Arts. 59, 73).

ii) Weaknesses

Screenshots and recordings are easy to **edit**, and **deepfakes** are a growing risk; the Supreme Court requires recordings to be proved **genuine and untampered** (*Ishtiaq Ahmed Mirza v Federation of Pakistan*, **PLD 2019 SC 675**), but many courts lack **forensic** support.

10. Challenges in enforcement

i) Unfinished legal business

The stamp-duty and attestation exemptions are **open-ended** (ss. 10–11), and **negotiable instruments, powers of attorney, wills and land contracts** remain excluded (s. 31).

ii) Capacity and awareness

The Council depends on **Government grants** and small fees (s. 20); police, prosecutors and judges need **digital-forensic** skills; and public awareness of accredited signatures is low.

iii) Cross-border dealings

The Ordinance reaches foreign acts (s. 32), but investigating and prosecuting **foreign** wrongdoers depends on international cooperation.

11. Reforms for the digital economy

i) Legal

Low-cost accredited signatures (for example, linked to national digital identity); completion of **e-stamping** and **e-attestation**; extension of the Ordinance to suitable excluded instruments (s. 31(2)); a **data-protection** law; and **harmonisation** of offences with PECA.

ii) Institutional

Funding and publicity for the **Council**; **forensic laboratories** and judicial training; and use of the **Digital Nation Pakistan Act 2025**, which created a **National Digital Commission** and a **Pakistan Digital Authority**, to integrate trusted digital identity and e-signatures into government services.

12. Conclusion — a sound framework awaiting implementation

i) Summary

The Ordinance **recognises** electronic records (ss. 3–6) and **signatures** (ss. 7–9), governs **contracts and communications** (ss. 13–15), builds a **certification system** (ss. 17–27) and punishes **misuse** (ss. 34–39), while amending evidence law. It secures **authenticity** and **reliability** well in law and **security** in part.

ii) The verdict

Its weakness lies not in its **text** but in **implementation** — adoption of accredited signatures, completion of electronic formalities, complementary data-protection and cyber laws, and forensic capacity. With Pakistan's digital economy growing, **making the existing framework work** is the priority.

Why this answer scores

- All four parts — provisions, certification, evaluation, challenges — are answered in order.
- Security, authenticity and reliability are each evaluated with strengths and weaknesses.
- The table gives the evaluation at a glance.
- Evidence includes PECA, the data-protection gap, NIFT, the Supreme Court and the Digital Nation Act 2025.
- Reforms are specific and tied to the digital economy.

Answer 6 · The Presumptions of Authenticity and Integrity — A Critical Examination (CSS 2026)

AT A GLANCE · CSS 2026 · Q6 · 20 MARKS

Critically examine the presumptions provided under the Electronic Transactions Ordinance, 2002 regarding the authenticity and integrity of electronic records and signatures. Analyze how these presumptions facilitate the reliability of electronic transactions and the evidentiary challenges that may arise in court.

The answer map

Section 9 presumes that a document bearing an advanced electronic signature is authentic, was signed with intention and has not been altered, unless contrary evidence is adduced; the Qanun-e-Shahadat Order adds presumptions that automated systems were working and makes printouts primary evidence. These presumptions make electronic transactions reliable by shifting the burden to the party denying them and rewarding secure technology. But compromised keys, low use of advanced signatures, unclear rebuttal standards, tampering and deepfakes, and limited forensic capacity create real evidentiary challenges.



Figure 8 · The answer map: every branch is one numbered heading of the model answer below.

Decoding the question

Three demands: (1) "**critically examine the presumptions**" — s. 9 of the Ordinance and the QSO presumptions (Art. 73, Expl. 3), with their conditions and limits; (2) "**analyze how**" they make transactions **reliable**; (3) the **evidentiary challenges** in court. Give each roughly a third, with examples and cases, and end with a judgment.

The model answer

1. Introduction — presumptions as the engine of digital trust

i) Definition

"In any proceedings, involving an **advanced electronic signature**, it shall be **presumed unless evidence to contrary is adduced**, that … the electronic document … is **authentic and has integrity**" (**s. 9**, Electronic Transactions Ordinance 2002). "In the absence of evidence to the contrary, it shall be **presumed** that the **automated information system was in working order**" (**QSO Art. 73, Explanation 3**).

ii) The argument

These presumptions make electronic transactions **dependable** in court by placing the burden on the party who **denies** them, and they **reward** secure, accredited technology. But they are **narrow** — available only for advanced signatures and automated systems — and they raise real **evidentiary challenges**: compromised keys, unclear rebuttal standards, tampering and limited forensic capacity.

2. Authenticity and integrity defined

i) Authenticity

"In relation to an electronic document or electronic signature, the **identification of and attribution to a particular person or information system**" (s. 2(1)(f)) — **who** made it.

ii) Integrity

The document or signature "has **not been tampered with, altered or modified** since a particular point in time" (s. 2(1)(q)) — **whether** it has changed.

3. The presumption for advanced signatures — s. 9

i) Limb (a)

The electronic document "affixed with an advanced electronic signature, as is the subject-matter of or identified in a **valid accreditation certificate**", is **authentic** and has **integrity**.

ii) Limb (b)

The advanced signature is "the **signature of the person to whom it correlates**", it "was **affixed by that person with the intention of signing or approving**" the document, and the document "**has not been altered** since that point in time".

iii) Rebuttable

The presumption holds "**unless evidence to contrary is adduced**" — it is a **rebuttable** presumption of fact and law.

4. What counts as advanced — the gateway to the presumption

An **advanced** electronic signature is either (i) **unique** to the signer, **capable of identifying** him, created by means under his **sole control**, and attached so that **any subsequent change** is **detectable**; or (ii) provided by an **accredited** certification service provider and **accredited by the Council** as capable of establishing authenticity and integrity (s. 2(1)(d)). The presumption therefore depends on the **certification system** — accredited providers (s. 17), approved practice statements (s. 25) and the public **repository** (s. 23).

5. Presumptions in the Qanun-e-Shahadat Order

i) Working order

A **printout** of an automated information system is **primary evidence** if the system was working, and its working order is **presumed** absent contrary evidence (Art. 73, Expl. 3).

ii) Relevance and attribution

Records of an automated system in working order are **relevant facts** (Art. 46-A), and may be **attributed** to the person controlling the system (Art. 30, Explanation). **Repository certificates** are **public documents** (Art. 85(6)).

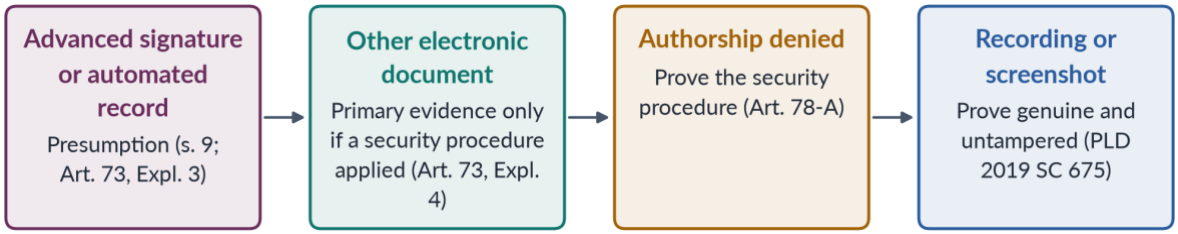


Figure 9 · Presumption or proof — the route in court.

6. Where no presumption applies — proof

i) Ordinary signatures

An ordinary electronic signature — typed name, scan, click — "**may be proved in any manner**" (s. 8), but receives **no presumption**.

ii) Denied documents

"Where such allegation is **denied**, the **application of a security procedure** to the signature or the electronic document **must be proved**" (Art. 78-A).

iii) Illustration

If Gulf Garments denies signing a revised contract, Sara Fabrics benefits from the presumption only if the contract bears Gulf's **SecureSign** advanced signature; if it bears only a typed name in an email, Sara must prove the **security procedure** — that it came from Gulf's password-protected account.

7. How the presumptions facilitate reliability

i) Shifting the burden

The party **denying** the signature — usually the one who knows of any misuse — must produce **contrary evidence**, reducing the cost and delay of proving every document.

ii) Rewarding security

Because only **advanced** signatures earn the presumption, businesses have an incentive to use **certificate-based** signatures and accredited providers.

iii) Certainty for commerce

Banks and businesses can rely on **certified** records and **system printouts** without calling technical witnesses in every case — just as the **s. 118 presumptions** of the Negotiable Instruments Act make paper instruments acceptable.

8. Challenge — compromised keys and identity theft

If a signer's **private key**, token or password is **stolen**, the presumption works **against the true owner**, who must prove misuse. Without access to the provider's **audit logs** or device records, this can be very difficult.

Illustration

A fraudster steals the USB token holding Ms Sara's private key and signs a contract in her name. The court presumes the signature is hers (s. 9). She must show that the token was stolen and reported, that SecureSign **revoked** the certificate in the repository (s. 23(4)), or that the signing took place from an unfamiliar device or location.

9. Challenge — rebuttal and the burden

i) Unclear standard

The Ordinance does not say **how strong** the "evidence to contrary" must be — whether a credible doubt suffices or proof on a balance of probabilities is needed.

ii) Inequality

The **working-order** presumption (Art. 73) favours **banks and telecom companies** that control the systems; an individual can rarely prove that their computers malfunctioned.

10. Challenge — tampering, deepfakes and forensics

i) Unsigned evidence

Most real-world evidence — **WhatsApp messages, screenshots, emails, recordings** — carries no advanced signature and is easy to **edit** or fabricate; **deepfakes** can imitate voices and faces.

ii) Judicial response

The Supreme Court has held that no audio or video can be relied on until proved **genuine and not tampered with**, and that forensic examination can establish this (*Ishtiaq Ahmed Mirza v Federation of Pakistan*, **PLD 2019 SC 675**); tapes require transcripts, **voice identification**, and disclosure of **source and date** (*Asfandyar v Kamran*, **2016 SCMR 2084**). Many trial courts, however, lack quick access to **forensic** laboratories.

11. A critical appraisal

i) Strengths

The presumptions are **well designed**: rebuttable, confined to **secure** signatures and **working** systems, and integrated with evidence law.

ii) Weaknesses

Their practical reach is **small** because accredited signatures are rare outside banks; the **rebuttal standard** is unclear; and victims of **identity theft** and individuals facing institutions bear heavy burdens.

12. Conclusion — useful presumptions needing support

i) Summary

Section 9 presumes advanced signatures **authentic, intended and unaltered**; the QSO presumes automated systems **working** and makes their printouts **primary evidence**; other documents must be proved, with a **security procedure** where authorship is denied (s. 8; Arts. 73, 78-A).

ii) The verdict

The presumptions **facilitate reliability** by shifting the burden and rewarding security. To work fairly, they need **wider access** to accredited signatures, **audit trails** that help signers rebut misuse, clear **rebuttal standards** and stronger **forensic** capacity in the courts.

Why this answer scores

- Both presumptions — s. 9 and Art. 73 — are quoted and examined with their conditions.
- It shows how the presumptions create reliability, with the NIA s. 118 analogy.
- The evidentiary challenges are specific and supported by Supreme Court authority.
- The flow figure maps presumption versus proof.
- A balanced appraisal and reforms complete the critical examination.

Answer 7 · Electronic and Computer-Generated Evidence under the Qanun-e-Shahadat Order and Alamgir Khalid Chughtai v State (CSS 2026)

AT A GLANCE · CSS 2026 · Q8 · 20 MARKS

Critically analyze the admissibility of electronic and computer-generated evidence under the Qanun-e-Shahadat Order, 1984. How does the judgment in Alamgir Khalid Chughtai v. State (PLD 2009 Lahore 254) contribute to the evolving jurisprudence on digital evidence in Pakistan?

The answer map

Since 2002 the Qanun-e-Shahadat Order makes computer-generated records relevant and attributable to the system's controller, treats printouts as primary evidence where the system worked or a security procedure applied, requires proof of a security procedure when authorship is denied, admits expert opinion on integrity, and lets courts admit modern-device evidence under Article 164 (substituted in 2023). Alamgir Khalid Chughtai v State (PLD 2009 Lahore 254) recognised rampant cyber-crime and a distinct statutory criterion for such evidence and is cited on authentication; the Supreme Court later required recordings to be proved genuine and untampered.



Figure 10 · The answer map: every branch is one numbered heading of the model answer below.

Decoding the question

Two limbs: (1) "**Critically analyze the admissibility** of electronic and computer-generated evidence under the QSO" — the 2002 amendments (Arts. 2, 30, 46-A, 59, 73, 78-A, 85(6)) and **Article 164** (with the 2023

substitution), with strengths and gaps; (2) "**How does *Alamgir Khalid Chughtai* contribute**" — its holding and its place in the case law, linked to later Supreme Court decisions. Give the first limb about 60% and the second about 40%.

The model answer

1. Introduction — from paper evidence to digital evidence

i) Definition

"For the purposes of this Ordinance, the **Qanun-e-Shahadat Order, 1984** … shall be read subject to the **amendments specified in the Schedule**" (**s. 29**, Electronic Transactions Ordinance 2002). Under Article 46-A, "statements in the form of electronic documents **generated, received or recorded by an automated information system while it is in working order**, are **relevant facts**" (**QSO Art. 46-A**).

ii) The argument

The QSO of 1984 was designed for **paper documents** and **oral testimony**. The **2002 Ordinance** rewrote it for the digital age — making computer records **relevant**, printouts **primary evidence** and electronic signatures provable — and the **2023** substitution of **Article 164** extended it to modern platforms. The courts, beginning with cases such as *Alamgir Khalid Chughtai*, then built safeguards for **authenticity**.

2. The framework — the Order as amended in 2002

i) Definitions

"Automated", "electronic", "information", "information system", "electronic document", "electronic signature", "advanced electronic signature" and "security procedure" bear the Ordinance's meanings in the QSO; "certificate" includes the Ordinance's meaning (Art. 2(1)(e)–(f)).

ii) Overall scheme

The amendments address **relevance** (Arts. 30, 46-A), **proof of contents** (Art. 73), **proof of authorship** (Art. 78-A), **expert opinion** (Art. 59) and **public documents** (Art. 85(6)).

3. Relevance of computer-generated evidence

i) Article 46-A

Records generated, received or recorded by an **automated information system** "while it is **in working order**" are **relevant facts** — bank statements, ATM logs, call-data records, server logs.

ii) Article 30, Explanation

"Statements generated by automated information systems may be **attributed to the person exercising power or control** over the said information system" — answering the hearsay problem of who "made" a machine's statement.

4. Primary evidence — printouts under Article 73

i) Explanation 3

A **printout** of an automated information system "shall not be denied the status of **primary evidence** solely" because it is electronic, if the system was working — and its working order is "**presumed**" absent contrary evidence.

ii) Explanation 4

A printout of any **other** electronic document "shall be treated as **primary evidence** where a **security procedure** was applied" when it was generated, sent, received or stored.

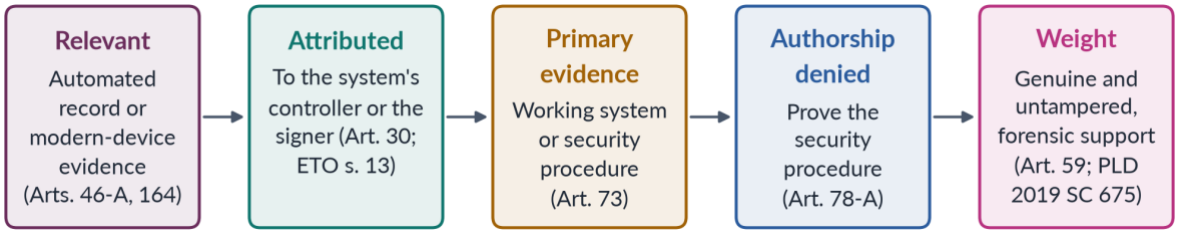


Figure 11 · Admitting electronic evidence under the QSO.

5. Proof of electronic documents and signatures

i) Article 78-A

"If an electronic document is alleged to be signed or to have been generated … by any person through the use of an information system, and where such allegation is **denied**, the **application of a security procedure** … **must be proved**."

ii) The Ordinance's presumption

An **advanced electronic signature** is presumed authentic, intended and unaltered unless contrary evidence is adduced (ETO s. 9); other electronic signatures may be **proved in any manner** (ETO s. 8).

6. Expert evidence and public documents

i) Article 59

Expert opinion is relevant "as to **authenticity and integrity** of electronic documents" and "as to the **functioning, specifications, programming and operations** of information systems" — the gateway for **digital forensics**.

ii) Article 85(6)

Certificates deposited in the repository under the Ordinance are **public documents**, provable by certified copies.

7. Article 164 — modern devices before and after 2023

i) The original text

The court could, "in such cases as [it] may consider appropriate", allow production of evidence "that may have become available because of **modern devices or techniques**" — the basis for admitting audio and video recordings.

ii) The 2023 text

The **Criminal Laws (Amendment) Act 2023** substituted Article 164: "Depending on the nature of case and circumstances, the Court may, if deem appropriate, allow to be produced any evidence or witnesses recorded by the Court through the modern devices or techniques including **video call, viber, skype, imo, whatsapp, facebook messenger, line caller and video conference**, etc."

iii) Its limits

Article 164 is **discretionary** and does not dispense with **proof**: the evidence must still be shown to be genuine.

8. Alamgir Khalid Chughtai v State

✦ LEADING CASE

Alamgir Khalid Chughtai v State · *Lahore High Court, PLD 2009 Lahore 254*

The Lahore High Court observed that **cyber-crime had become rampant** in society and that, because electronic communications connect people across the world without any physical link, the **legislature had provided a different criterion for the admissibility of evidence** in such cases. Later judgments cite it on the authentication of e-mails and electronic texts — including that authorship may be established where the **adverse party admits** writing them. (The full facts of the case could not be verified for these notes.)

9. Its contribution to the jurisprudence

i) A distinct statutory regime

The case recognised that electronic evidence is governed by **its own statutory criteria** — the 2002 amendments to the QSO — rather than by rules made for paper and oral evidence.

ii) Judicial awareness

By acknowledging that cyber-crime was **rampant**, the Court signalled that judges must adapt their approach to **proof** in the digital age.

iii) Authentication

It is cited in later decisions on how a trial judge can be satisfied that an electronic text was really written by a party — **admission**, surrounding **circumstances**, and proof of the **account or system** used — complementing Art. 78-A.

iv) A foundation

It stands at the start of a line of authority that the **Supreme Court** later developed into detailed conditions for audio, video and digital evidence.

10. Later developments — Asfandyar and Ishtiaq Ahmed Mirza

i) Asfandyar v Kamran (2016 SCMR 2084)

Audio or video tapes cannot be relied upon unless **genuine and unaltered**; a **transcript** should be prepared under independent supervision; the **voice** must be identified; the **source** and **date** of acquisition disclosed; and a **formal application** filed.

ii) Ishtiaq Ahmed Mirza v Federation of Pakistan (PLD 2019 SC 675)

"**No audio tape or video** can be relied upon by a court until the same is proved to be **genuine and not tampered with or doctored**"; Article 164 is **discretionary**, and even admitted recordings must be proved under the law of evidence; **forensic** examination can establish genuineness, and a **Punjab Forensic Science Agency** report is per se admissible.

11. Critical analysis — strengths and gaps

i) Strengths

Pakistan's framework **admits** digital evidence on clear statutory terms, gives **printouts** primary status, uses **security procedures** and **certificates** to establish authorship, and allows **forensic** experts — balanced by Supreme Court safeguards.

ii) Gaps

Deepfakes and edited screenshots threaten authenticity; there are no detailed statutory rules on **chain of custody** outside PECA; the **working-order** presumption favours institutions; many courts lack **forensic laboratories**; and Article 164's wide discretion can produce **inconsistent** practice.

12. Conclusion — admissible but proved

i) Summary

Under the QSO as amended in 2002, computer records are **relevant** and **attributable** (Arts. 30, 46-A), printouts are **primary evidence** (Art. 73), denied documents require proof of a **security procedure** (Art. 78-A), experts may testify (Art. 59), and Article 164 (2023) admits modern-device evidence. *Alamgir Khalid Chughtai* recognised a **distinct criterion** for such evidence and is cited on **authentication**; the Supreme Court has since required recordings to be proved **genuine and untampered**.

ii) The verdict

Electronic evidence is **admissible**, but its **weight** depends on proof of authenticity and integrity. Forensic capacity, chain-of-custody rules and judicial training are the next steps.

Why this answer scores

- It sets out every QSO amendment relevant to electronic evidence, with exact words.
- Article 164 is explained in both its original and its 2023 form.
- *Alamgir Khalid Chughtai* is stated accurately, without inventing facts, and its contribution is analysed.
- Later Supreme Court cases show how the jurisprudence evolved.
- The critical analysis identifies concrete strengths and gaps.